2.1 STUDENT'S *t* DISTRIBUTION AND ITS APPLICATIONS

2.1.1 Student's *t*-distribution

χ^2-distribution and some of its applications are discussed in Section 2.2

If $X \sim N(0,1)$ and $Y \sim \chi^2_n$ are independent random variables, then

$T = \dfrac{X}{\sqrt{Y/n}}$ is said to have *t*-distribution with *n* degrees of freedom. This can be denoted by t_n.

Note 1: The degrees of freedom of *t* is the same as the degrees of freedom of the corresponding chi-square random variable.

Note 2: The *t*-distribution is used as the sampling distribution(s) of the statistics(s) defined based on random sample(s) drawn from normal population(s).

i) If $X_1, X_2, \ldots, X_n$ is a random sample drawn from $N(\mu, \sigma^2)$ population then

$$X = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \text{ and } Y = \frac{\sum (X_i - X)^{\sim}}{\sigma^2} \sim \chi^2_{n-1} \text{ are independent.}$$

Hence,

$$T_1 = \frac{X}{\sqrt{Y/n-1}}$$

$$= \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \times \frac{\sigma\sqrt{(n-1)}}{\sqrt{\sum (X_i - \overline{X})^2}}$$

$$= \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1} \qquad \text{where } S = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{(n-1)}}.$$

ii) If $(X_1, X_2, \ldots, X_m)$ and $(Y_1, Y_2, \ldots, Y_n)$ are independent random samples drawn from $N(\mu_X, \sigma^2)$ and $N(\mu_Y, \sigma^2)$ populations respectively, then

$$\frac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{\sigma\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim N(0,1) \text{ and } \frac{\sum_{i=1}^{m} (X_i - \overline{X})^2 + \sum_{j=1}^{n} (Y_j - \overline{Y})^2}{\sigma^2} \sim \chi^2_{m+n-2} \text{ are independent.}$$

Then,

$$T_2 = \frac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}$$

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